Jared Narvaez

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EDUCATION

University of California, Berkeley Expected Graduation: **May 2027**
Bachelor's Degree in Computer Science; Minor in Data Science
Relevant Coursework: Data Structures, Machine Structures, Computer Security, Computer Networking, Artificial Intelligence

EXPERIENCE

Multi-Agent Systems & Agentic Product Engineer Intern *October 2026 – Present*
PhysaFlow *San Francisco, CA (Remote)*

- Designed a multi-agent decision layer for data-center operations with three deterministic domain agents (facility power/cooling, compute/IT health, carbon) that can only propose changes or decline with a stated reason, never issue commands or write to systems.
- Implemented the agent package in Python 3.11 with typed proposal/no-op contracts, a shared response envelope, pytest fixtures, and an import-allowlist test that fails CI if an agent depends on anything beyond its validated input contract.
- Wrote the design doc and delivery plan, raising 10 interface-contract questions that the tech lead resolved into 3 contract changes I own, delivered through branch-and-PR code review with green CI.
- Build fail-closed input handling that turns missing, stale, or invalid data into an explicit labeled state instead of a hard failure.
- Validate each agent end to end against a coordinator and an independent safety supervisor across 4 scenarios: nominal, conflict, missing data, and source failure.

Data Assistant *June 2026 – Present*
UC Berkeley, Visiting Scholar & Postdoc Affairs (VSPA) *Berkeley, CA*

- Debugged and wrote Python scripts for data validation and cleaning, standardizing 3,000+ monthly entries from departments and funding agencies.
- Produce monthly appointment and reappointment summaries from the validated data, giving the Vice Chancellor's office audit-ready statistics across 90+ departments and 3 scholar types each quarter.

Core 38, 39 Vice President of External Relations *May 2025 – Present*
Pilipinx Association of Scientists, Architects & Engineers (PASAE) *Berkeley, CA*

- Led semester recruitment pipeline (tabling, info sessions, kickoffs) and co-organized UC Berkeley career fairs hosting 90+ employers and 2,000+ students; secured speakers and sponsors for flagship events.
- Managed and mentored External Reps and interns; ran weekly standups, scoped workstreams, created onboarding playbooks, and served as a career resource for 60+ active members.

PROJECTS

Secure File Storage & Sharing System (Go) | *Course Project* *Fall 2025*

- Designed and built a secure cloud file storage system in Go, using Argon2 key derivation, AES-CTR encryption, and HMAC-based integrity verification to protect user credentials and file contents against malicious server operators and unauthorized access.
- Engineered an append-efficient linked-list storage architecture with per-chunk encryption, achieving O(new data) bandwidth on appends; file keys distributed via RSA public-key encryption and authenticated with digital signatures.
- Implemented invitation-based file sharing with transitive access and secure revocation, re-encrypting content under fresh keys and recursively propagating updates across a sharing tree; supported multi-device sessions with a fully stateless client.

Reliable Transport Protocol (TCP) (Python) | *Course Project* *Spring 2026*

- Implemented a subset of TCP with a full connection state machine (11 states), including three-way handshake, bidirectional data transfer, and graceful shutdown via both active and passive close procedures.
- Built reliable in-order delivery over an unreliable channel using a receive queue for out-of-order packets, sliding window flow control with advertised window updates, and sequence number wraparound handling.
- Engineered a retransmission system with a retransmit queue and adaptive timeout, estimating RTT via RFC 6298's exponential moving average algorithm and dynamically adjusting RTO to achieve within 5–10% of true RTT under varying link latencies.

SKILLS

Languages: Python, Go, Java, C, SQL, JavaScript, HTML/CSS
Frameworks & Tools: pytest, GitHub Actions (CI), Git/GitHub, React, Node.js, NumPy, Pandas, Wireshark
Concepts: Multi-Agent Systems, Agent Evaluation, LLMs, Prompt Engineering, Software Testing, Cryptography, Networking (TCP/IP)